

3. $f: A \rightarrow B$ ~~ex. 1~~ bijection: surjection + injection

Def. Let $V, W \in \mathcal{V}$. Then V and W are isomorphic (同构) if

$\exists \sigma \in \text{Hom}(V, W)$ s.t. σ is an isomorphism i.e. σ is a bijection

eg. $\mathbb{R}^2$ and $\mathbb{R}$ are isomorphic as sets

but $\forall \sigma \in \text{Hom}(\mathbb{R}^2, \mathbb{R})$ is not an isomorphism (as v.s.)

$\hookrightarrow$ injection: $\exists x \neq 0 \in \mathbb{R}^2$ s.t. $\sigma(x) = 0$

Ex. If $\mathbb{F}^n \xrightarrow{\sigma} \mathbb{F}^m$ is surjective $\Rightarrow n \geq m$

$$\begin{aligned} (x_i)_{i=1}^n = x &\mapsto \sigma(x) = \sigma\left(\sum x_i e_i\right) = \sum x_i \sigma(e_i) & p_i = \sigma(e_i) = \begin{pmatrix} a_{1i} \\ a_{2i} \\ \vdots \\ a_{mi} \end{pmatrix} \\ &= \sum x_i p_i \\ &= \begin{pmatrix} p_1 & p_2 & \dots & p_n \end{pmatrix} x \end{aligned}$$

1. σ is surjective $\Leftrightarrow \forall y. \exists x$ s.t. $y = Ax \Leftrightarrow \forall y \in \mathbb{F}^m \Rightarrow y \in \mathcal{R}(A)$

$\Leftrightarrow \mathcal{R}(A) = \mathbb{F}^m. \Leftrightarrow r(A) = m \Leftrightarrow n \geq m$ (full row rank)

2. σ is injective $\Leftrightarrow$ full column rank

Thm1. Let $\sigma \in \text{Hom}_{\mathbb{F}}(\mathbb{F}^n, \mathbb{F}^m)$ be an isom. Then $n = m$ (反之亦然)

Thm2 (Generalization of Thm1) Let V and W be finite dimensional vector spaces over $\mathbb{F}$. Then $V \cong W \Leftrightarrow \dim V = \dim W \Leftrightarrow V \cong \mathbb{F}^{\dim V}$

Pf. " $\Rightarrow$ " Let $\sigma \in \text{Hom}(V, W)$ be an isomorphism

Let $\mathcal{B}_V = \{v_1, v_2, \dots, v_n\}$ be a basis of V , claim $\{\sigma(v_1), \dots, \sigma(v_n)\}$ is a basis of W

@ 1. Let $x_1 \sigma(v_1) + x_2 \sigma(v_2) + \dots + x_n \sigma(v_n) = 0$

$$\text{Then } \sigma(x_1 v_1 + x_2 v_2 + \dots + x_n v_n) = 0$$

Γ σ is injective $\Leftrightarrow \sigma(v) = 0 \Rightarrow v = 0$

kernel 核 $\ker \sigma = \{v \in V \mid \sigma(v) = 0\}$

Properties 1 σ is injective $\Leftrightarrow \ker \sigma = 0$

σ is inj $\Rightarrow x_1 v_1 + x_2 v_2 + \dots + x_n v_n = 0$, so $x_1 = x_2 = \dots = x_n = 0$

@ maximal $\forall w \in W, \exists v \in V$ s.t. $\sigma(v) = w$ (surjective)

" $\Leftarrow$ " Let $\dim V = \dim W = n$

$$v_i \mapsto \sigma(v_i) = w_i, 1 \leq i \leq n$$

$$v = \sum x_i v_i \xrightarrow{\text{linear expansion}} \sigma(v) = \sum x_i \sigma(v_i) = \sum x_i w_i$$

① σ is injective suppose $\sigma(v) = 0$ i.e. $\sum x_i w_i = 0 \Rightarrow x_1 = x_2 = \dots = x_n = 0 \Rightarrow v = 0$

② σ is surjective $w = \sum x_i w_i = \sum x_i \sigma(v_i) = \sigma(v)$

Ex. $F = \mathbb{R} \quad V = \{y \mid y'' - 2y' + y = 0\} \quad W = \{f(x) \in \mathbb{R}[x] \mid f \in \mathcal{Z}, f(1) = f(2) = 0\}$

so $\dim V = 2 \quad \mathcal{B}_V = \{e^x, xe^x\}$

$\dim W = 2, \mathcal{B}_W = \{x(x-1), x^2(x-1)\}$

construct an isomorphism $\sigma: V \xrightarrow{\sim} W$

$$\Rightarrow e^x \mapsto x(x-1) = x(x-1) + 0 \cdot x^2(x-1) = (w_1, w_2) \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$xe^x \mapsto x^2(x-1) = 0 \cdot x(x-1) + x^2(x-1) = (w_1, w_2) \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$c_1 e^x + c_2 x e^x = y \mapsto \sigma(y) = c_1 \overset{w_1}{x(x-1)} + c_2 \overset{w_2}{x^2(x-1)} = (w_1, w_2) \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$

properties 2. σ is surjective $\Leftrightarrow \text{Im } \sigma = \{w \in W \mid \exists v \in V, \text{ s.t. } w = \sigma(v)\}$

3. Fix a basis $\{v_1, v_2, \dots, v_n\} = \mathcal{B}_V$ resp $\mathcal{B}_W = \{w_1, w_2, \dots, w_m\}$

suppose $\sigma(v_i) = p_{i1}w_1 + p_{i2}w_2 + \dots + p_{im}w_m \quad (1 \leq i \leq n)$

$$= (w_1, \dots, w_m) \begin{pmatrix} p_{11} \\ \vdots \\ p_{ni} \end{pmatrix}$$

$P = (p_{ij})_{m \times n} \in F^{m \times n} \rightarrow$ transition matrix 过渡矩阵.

$$v \in V \xrightarrow[A_{n \times n}]{\sigma} W = \sigma(v)$$

$$\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in F^n \xrightarrow[A_{m \times n}]{P} F^m$$